14. Bill of Materials (BOM)

Part NO.	Part Name (Make)	Material specification	Qty.	Technical Specification	Position in Technical assembly
1	Egg Storage Container	PP-C HI / Polypropylene Copolymer	1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	1
2	Egg Tunnel	PP-C HI / Polypropylene Copolymer	1	Yield: ~22 MPa | Density: 0.90 g/cm³ | Temp: -20 to 100°C	2
3	Gates for Transportation Tunnel	Acetal Copolymer (POM-C)	2	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	4
4	Gate Shaft Holder for Tunnel	POM-C / Acetal Copolymer	2	Yield: ~62 MPa | Density: 1.41 g/cm³ | Temp: -40 to 100°C	5
5	Egg Tunnel Shaft	EN 1.4301 / AISI 304	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	6
6	Egg Holding Shaft	EN AW-6082 T6	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	7
7	Egg Holding Arm	EN AW-6082 T6	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	8
8	Base of Spear	EN AW-6082 T6 (clear anodised)	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	9

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9	Spears (Left and Right)	EN 1.4116 / X50CrMoV15 (hardened)	2	Yield: ~800–1000 MPa | Density: 7.70 g/cm³ | Temp: -20 to 200°C	10
10	Solenoid Motor Mounting Bracket	EN 1.4301 / AISI 304	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	11
11	Solenoid Motor Casing	EN AW-6082 T6 (clear anodised)	3	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	12
12	Main Drawer Box	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	13
13	Drawer Front Cover	ABS-HI plastic	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	14
14	Drawer Handle	ABS-HI plastic w/ brass inserts	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	15
15	Inner Shaft (Drawer)	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	16
16	Manual Compression Plate	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	17
17	Oil Storage Mount	EN AW-6082 T6	2	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	18
18	Pocket Plate	EN 1.4409 / GX2CrNiMo19-11-2 (cast)	4	Yield: ~185–205 MPa | Density: 7.90 g/cm³ | Temp: -196 to 400°C	19

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19	Sprocket for Conveyor Belt	EN 1.4404 / X2CrNiMo17-12-2	4	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	20
20	Driving Shaft / Idler Shaft	EN 1.4404 / X2CrNiMo17-12-2	2	Yield: ~205 MPa | Density: 8.00 g/cm³ | Temp: -196 to 400°C	21
21	Conveyor Side Frame	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	22
22	Conveyor Back Frame	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	23
23	Conveyor Belt Hollow Support Frame	EN AW-6082 T6	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	24
24	Nema 17/Gearbox/Encoder Motor Housing	EN AC-43500 / AlSi10MnMg (cast)	1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	25
25	Nema Motor Mount	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	26
26	TOP IR Shielding	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C (shield, near IR zone)	27
27	Ceramic Block for Holding Quartz IR	96% Al_2O_3 ceramic	2	Flexural: ~300 MPa | Density: 3.70 g/cm³ | Temp: up to 1600°C	28
28	Side Lining for TOP IR	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	29

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29	Perforated Sheet	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	30
30	TOP IR Back Frame	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	31
31	Heating Block	EN AW-6082 T651	1	Yield: ~250 MPa | Density: 2.70 g/cm³ | Temp: -40 to 200°C	32
32	Side Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	33
33	Front Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10	2	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	34
34	Bottom Frame (Bottom Heating)	EN 1.4301 / X5CrNi18-10	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	35
35	Mounting Brackets for Bottom Heating Block	EN 1.4301 / AISI 304	4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	36
36	Aluminum Plate (Rotating Plate)	EN AW-6061 T6 (clear anodised)	1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	37

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37	Top Load Cell Plate	EN AW-6061 T6 (clear anodised)	1	Yield: ~275 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	38
38	Bottom Load Cell Plate	EN AW-6082 T6 (clear anodised)	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	39
39	Base Plate (Rotating Plate)	EN AW-6082 T6 (clear anodised)	1	Yield: ~260 MPa | Density: 2.70 g/cm³ | Temp: -40 to 150°C	40
40	Stepper Motor Housing	EN AC-43500 / AlSi10MnMg (cast, clear anodised)	1	Yield: ~140–180 MPa | Density: 2.68 g/cm³ | Temp: -40 to 150°C	41
41	Bottom Base for Rotating Plate	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	42
42	Electrical Box Housing	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	43
43	Main Frame Housing	EN 1.4301 / AISI 304	1	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	44

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44	L Bracket for Servo Motor	EN 1.4301 / AISI 304	4	Yield: ~215 MPa | Density: 7.90 g/cm³ | Temp: -200 to 400°C	45
45	Plastic Back Plate (Display)	ABS-HI plastic w/ brass inserts	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	46
46	Front Frame for Casing (Display)	ABS-HI plastic	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	47
47	Main Case/Chassy Panels	ABS-HI plastic + purchased glass	1	Yield: ~40 MPa | Density: 1.05 g/cm³ | Temp: -20 to 80°C	48

Part NO.	Part Name (buy)	Material specification	Link/Brand	Technical Specification	Position in Technical assembly
48	DIN Rail	zinc-plated steel	Hutschiene TS 35 x 7,5mm gelocht - L=326mm	35 x 7,5 x 326 mm	49
49	Cable Duct / Wire Raceway	PVC	Cable duct	25x25x1000mm	NA
50	Cable Glands / Strain Reliefs	Polyamide, V-2.	Kabelverschraubung, M 12 x 1,5, Ø 3,5 - 7 mm, schwarz, IP69K Kabelverschraubung, M 16 x 1,5, Ø 4,5 - 10 mm, schwarz, IP69K M 20 x 1,5, Ø 7 - 13 mm, schwarz, IP69K	M12x1.5 IP69k M16x1.5 IP69k M20x1.5 IP69K	NA
51	Cable Ties + Mounts	6/6 nylon	Cable mounts Cable ties	18.8mm x 18.8mm x 5.6mm	NA

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52	Axle Shaft	Aluminum	Aluminum shaft	Ø 10 mm x100 mm	50
53	Return Spring	Stainless steel - AISI 302	Spring	L0 - 20.57 Mm De - 7.62	NA
54	Thermal Insulation	LDPE	Thermal insulation	-70°C to105°C 500x330x20mm	NA
55	Inner Lining	Aluminum	Aluminum sheet	0.1x100x100mm	NA
56	Inline Check Valve	stainless steel	Duotight - 8mm Inline Check Valve	Ø8mm	NA
57	Silicone Tubing	Silicon	Silicon tubing	4mm x 1m	NA
58	Flat Fan Spray Nozzle	stainless steel	CDIYTOOL 8 Pieces 65 Degree High Pressure Spray Nozzle	0.4-5L/min 14mm hex width, 25mm long, 12.5 thread width	NA
59	Nozzle Mounting Bracket				NA
60	Roller Chain with Bent Attachments	Steel	IWIS - Rollenkette EuroChain	9,525 mm 5,72 mm 1505mm 06B-2	NA
61	2-Bolt Flange Bearing Block	Aluminum	Stehlagereinheiten / Flanschlagereinheiten 2- Loch KFL000	Ø 10mm	51
62	Flexible Clamp Motor Coupling	Aluminum	10 mm to 12 mm Flexible Shaft Coupling, Diameter 40 mm Length 66 mm Aluminium Motor Connector Flexible	bore diameter 10 mm to 12 mm total diameter 40 mm length 66mm	NA

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63	Planetary Gearbox for Motor	--	Planetengetriebe Schrittmotor 27:1	Planetarygear 27:1	52
64	Front Sealing Strip / Gap Cover	Silicon	EKI 1515 Silikonschaum	2mmx6mmx10m -60 °C bis +220 °C	NA
65	Thermal Compound	--	Wärmeleitpaste, Tube, 35gr, ungiftig	35g	NA
48	Stainless Fasteners Kit	A2 stainless steel	M2 M3 M4 M5 Schrauben Set – 1280 Stück	M2 M3 M4 M5	NA
Miscellanous					
67	Countersunk Raised Head Cross Recess Screw	A2 Stainless Steel	Senkkopfschrauben M10x25 Senkkopfschrauben M10x60	M10x25 M10x60	53
68	Pan Head Cross	A2 Stainless steel	Linsenschrauben M8x16 STM390400300S STM320300200E	DIN 7985 (ISO 7045) M8x16 M4x30 M3x20	54
69	Recess Screw	A2 stainless steel	Zylinderschraube Innensechskant DIN912 (ISO 4762) V2A - M6x20, 100 St. Zylinderschraube Innensechskant DIN912 (ISO 4762) V2A - M3x12, 100 St.	M6x20 M3x12	55
70	Socket Head Cap Screw	A2 stainless steel	DIN 912/ISO 4762 A2 – M3 x 25 mm 100 Stück DIN 912/ISO 4762 A2 – M3 x 10 mm 100 Stück	M3x25 M3 x 10 mm	56

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71	Pan Head Cross Recess Screw	Stainless steel	Flachkopfschraube ISO 7045 PZ M4x8-Z 4.8 galv.verz. 2000St.	ISO 7045 M4x8 M4 x 8mm	NA
72	Prevailing Torque Hex Nut	A2 Stainless steel	Sechskantmuttern M6 DIN 934 Edelstahl A2	M6	57
73	Plain Washer	A2 Stainless steel	Unterlegscheibe DIN125 (ISO 7089) V2A - 6,4 mm (M6), 1000 Stück	M6	58
74	Socket Set Screw, Cup Point	A2 Stainless Steel	Gewindestift DIN 916 / ISO 4029, M 5x20, Stahl 45 H	M5x20	59
75	Hex Thin Nut	A2 Stainless Steel	DIN 439/ISO 4035 Sechskantmutter Edelstahl A2 – M3 - 100 Stück DIN 439/ISO 4035 Sechskantmutter Edelstahl A2 – M4 - 100 Stück	M3 M4	60
76	Lazy Susan Bearing	Stainless Steel	Blickle 942239 DK-L-3 Stahlblech-Drehkranz 1 St.	140 x 110 mm	61
77	Rubber Drive Wheel	Polyoxymethylene	Freies Verschiffen 10PCS 8*32*16 flache typ PA nylon	8x32x16mm	62

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Electrical Parts

Part No.	Part name	Brand/link	Technical specifications	Location in technical assembly
78	IEC C20 Power Inlet	IEC 20 Power Inlet	16 A 250 VAC	63
79	MCB-Circuit Breaker	16A MCB- TYPE B Circuit Breaker	16A TYPE B	NA
80	RCCB / RCD-TYPE A	RCCB / RCD-TYPE A (30mA)	30mA TYPE A	NA
81	DIN Terminal Blocks	DIN Terminal Blocks (L/N/PE)	(L/N/PE)	NA
82	Protective Earth Terminal Block	Protective Earth Terminal Block	-	NA
83	AC Wire	LAPP 70261141 Strom Anschlusskabel Schwarz 3.00 m	3.00 m	NA
84	24V Industrial SMPS	24V Industrial SMPS 60W (2.5A at 24V)	60W (2.5A at 24V)	NA
85	24V → 12V LM2596T-12	LM2596T-12/LF03	4.5V to 40V 3A Low	NA
86	12V → 5V LM2596T-5	LM2596T-5.0/LF03	4.5V to 40V 3A Low	NA
87	24V → 6V LM2596S-ADJ (Adjusted to 6V)	LM2596S-ADJ/NOPB	24V DC to 6V DC, 2A continuous	NA
89	5V → 3.3V Regulator AMS1117-3.3	LM1117MPX-3.3/NOPB	5 mA 3.3 V -40 to 125 C	NA
90	DC Terminal Blocks	9-Way DC Terminal Block 5V 12V 24V for CCTV Power Supply	5V 12V 24V DC 9-way	NA
91	Inline Blade Fuse Holders	URAQT Fuse Holder Car Standard 16AWG	16AWG/14AWG/12AWG 30/20/10A	NA

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92	Blade Fuse Assortment	0287010.PXCN Littelfuse Inc. | Fuses | DigiKey	10A 32V	NA
93	Shielded Signal Cable	2M UL2547 28AWG 4 Cores PVC	81–600 V 4-core	NA
94	Ferrules and Crimp Terminals	1-966066-1 : Crimp Wire Pins, Tabs & Ferrules | TE Connectivity	15 AWG	NA
95	PCB Screw Terminal Connectors	20Paar 5.08mm Pitch Male Buchse PCB Screw Terminal Block | eBay.de	25 x 10 x 15mm AC 300V 15A	64
96	Axial fan 8414NGML	EBM Papst 8414NGML Axiallüfter 24 V/DC 42 m³/h (L x B x H) 80 x 80 x 25mm | » Freue dich auf voelkner	80x80x25mm 42 m^3/h 2pcs 24V DC	65
97	Main Microcontroller STM32H743ZITx	STM32H743ZIT6 STMICROELECTRONICS, ARM-MCU, STM32 Family	STM32 3.6V 550MHz	NA
98	Gate Actuators JF-0530B x3	JF-0530B DC12V 300mA 5N/10mm Pull-Push-Type Solenoid Electromagnet	DC 12V 300mA 5N/10mm 30x16x15mm	66
99	Flow Sensors IR sensor	IR Break Beam Sensors- 3mm LEDs : Adafruit Industries	3.3 - 5.5VDC 10mA @ 3.3V, 20mA @ 5V Response Time: <2 ms	NA
100	Flip Actuator MG996R	MG996R Digital Servo Motor mit Metall Getriebe	0.17 sec/60 deg at 4.8 V 4.8 to 7.2 V 40 x 19 x 43 mm	67
101	Platform Solenoid JF-0837B	JF-0837B DC 24V 1A 15N 10mm Push Pull Type Open Frame Linear Motion Solenoid Electromagnet | Harfington	DC 24V 1A 15N 37 x 20 x 26mm	68

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102	Solenoid Driver multi-channel MOSFET	Industrial-Grade MOSFET Driver Module – 10-100A Adjustable PWM Controller (12-60V DC)	12–60 V DC 10–100 A 12 x 13 x 1 cm	NA
103	Peristaltic Pump	DC 12V Dosing Pump Peristaltic Pump	DC12V 11ml/min Three roller	69
104	MOSFET Driver Module	Mosfet Mosfet 12V DC 8×6×2 Low Voltage Mosfet Module	12V DC 1-5MA	NA
105	Touchscreen Display Module — Nextion NX8048P070_011R	NEXTION 4.3/5.0/7.0" Intelligent LCD Touch Display	108x181x9.3mm	70
106	JST-XH 4-pin Connector	JST JST - Buchsengehäuse, 1x4-polig - XH	250V / 3A -25 to 80C	NA
107	Stepper Motor	NEMA 17	2.8 V 5 mm 2 phase	71
108	Stepper Motor Driver	Stepper Motor Driver	-40 to +85°C 12/24V 18 x 25 x 3 cm	NA
109	Control Wiring and Connectors	Control Wiring and Connectors	26AWG L-100 mm	NA
110	Hall Effect Sensor	A1104-class Hall effect switch sensor, 3-pin type	3.8 V to 24 V 25 mA -40 to 150 °C	NA
111	SmCo Magnet	SmCo	8 x 3 mm	NA
110	Incremental Rotary Encoder	AMT102-V	3,6V to 5,5V	72

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111	Cartridge Heaters	Cartridge Heaters	250W 230V (AC)	73
112	Temperature Sensor	PT100 temp. Sensor	-	74
113	Temperature Sensor Amplifier	MAX31865AAP+ Analog Devices DigiKey	3V to 3,6V -40°C to125°C	NA
114	Solid-state Relay	solid state relay 25A SSR-25AA , ZG3NC-325A Single phase	25A 90–480VAC	75
115	Heater Branch Fuse	0652C8000-11 Bel Fuse Inc. | Sicherungen | DigiKey	8 A 250 V	NA
116	High-temperature Wiring	Silikonkabel 18 AWG 0,5 mm² 200°C 300V flexibel verzinnt Meterware | akkuteile.de	18 AWG 200°C 600V	NA
117	Grounding Hardware	100m Erdkabel H07V-K 1x4mm² Grün/Gelb bei hd-digital kaufen,	AC Uo/U 450/750 V > 20 MΩ × km -30 °C to +80 °C	NA
118	IR Heater Element	IR HEATER	410 mm 250W 110V to 450V	76
119	IR Power Control / Relay	SSR Halbleiter Solid State Relais DC / AC, 230V AC 25 A	DC to AC 230V AC 25A	NA
120	Thermal Cutoff / Safety Sensor	THERMODISC Temperatursicherung, 16A, 184°C, kurz/lang | Temperatursicherungen günstig kaufen | reichelt elektronik	16A Max 184°C	NA
121	PT100 Sensor	PT100 temp. Sensor	-	77
122	SSR Heat Sink	Aluminium Heat Sink SSR	Aluminium	78

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		Dissipation For One Phase Solid State Relay 10a-40a For PID Thermostat Temperature Controller	Fit for: 10a-40a	
123	Ceramic Terminal Block	Elektrofachmarkt-online - Keramikklemme 2-polig mit Befestigungsloch	Ceramic clamp 2-pole max. 300°C , 450V, 32A	NA
124	Low-voltage Terminal Blocks	Phoenix Contact 1725672 MPT 0,5/ 4-2,54 Leiterpla	160 V Max 105 °C 4 cores	NA
125	Stepper Motor	FREI Standard-Schrittmotor, 5V	5V DC 4 Phase 34.3mN.m	NA
126	Stepper Driver Module	FREI Entwicklerboards - Motorsteuerung, ULN2003A	DC5V-12V 4-phase 5 line	NA
127	Microswitch	5Pcs Mini Micro Limit Switch NO NC 3 Pins PCB Terminals SPDT 5A 125V 250V 29mm Roller Arc lever Snap Action Push Microswitches	5A 125V/250V	NA
128	Load Cell and HX711 Amplifier Module	Stemedu 5 kg Load Cell Digital Weight Sensor for Portable Electronic with HX711 Ad Amplifier Module, 11 lbs Weighing Sensor Arduino Raspberry Pi	Four lines 5-10 V Dimensions: 12.7 mm, 12.7 mm, 80 mm. voltage range: 2.6-5.5 V	NA
129				
130	Wiring and Connectors	DKS-20MM-30 DigiKey Standard | Jumper Wire | DigiKey PVC Datenleitung LAPP UNITRONIC LiYY 3x0,25 - 0028303 | Automation24	-	NA
131	Emergency stop button	1NC 22mm Red Mushroom Emergency Stop Push Button Switch 600V 10A	22mm 600V 10A	NA

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